

Lecture 3

The Algebra of Scalars

→ this is easy! It's just the same math you have been doing your whole life!

Addition, subtraction, multiplication, division work exactly the same way!

BIG CAVEAT! • UNITS MATTER

✓ $m^2 = m_1^2 + m_2^2$ (m is mass)
(kg²) (kg²) (kg²)

✗ $m = 2m_1 + 3m_2$
(kg) (kg) (kg²)

1 1 1 ... mass of vectors

The Algebra

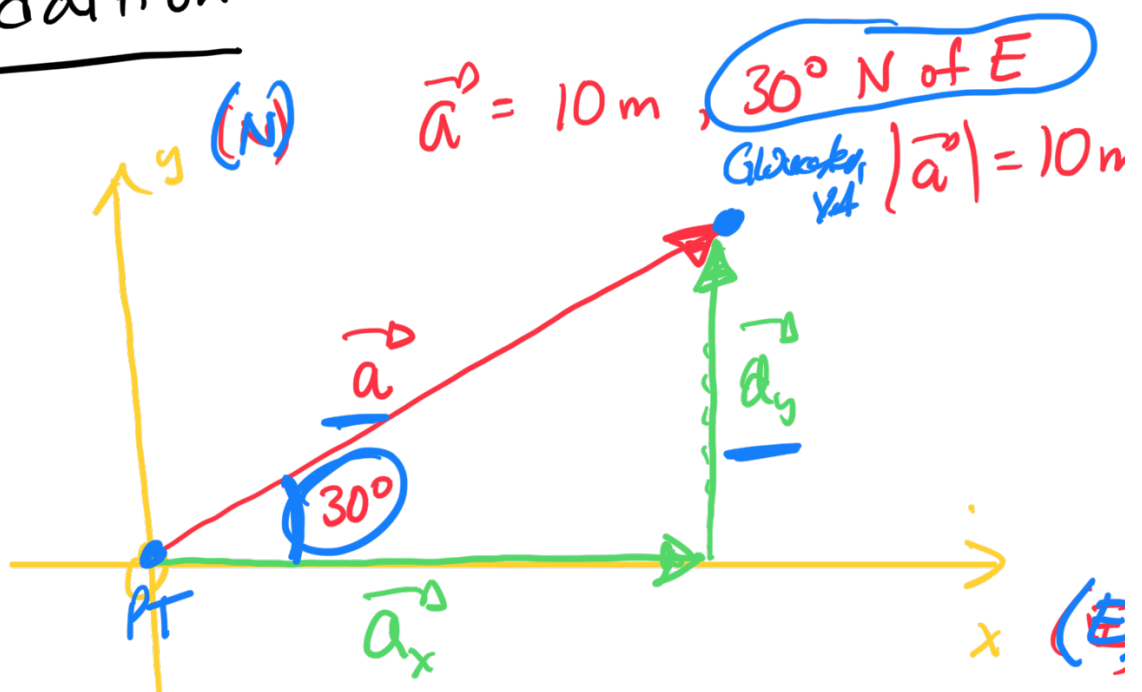
→ This is much different!
→ Direction Matter!!

$$a = 2\text{m} + 3\text{m} \quad \therefore a = 5\text{m}$$

$$\vec{b} = 2\text{m South} + 3\text{m North}$$

= ?

Addition



$$\vec{a} = \vec{a}_x + \vec{a}_y$$

i.e. We split the vector $\vec{a}$ into an x-component and a y-component

Step 1: Pythagorean Theorem.

$$|\vec{a}|^2 = |\vec{a}_x|^2 + |\vec{a}_y|^2$$

$$100 = (10\text{m})^2 = |\vec{a}_x|^2 + |\vec{a}_y|^2$$

Step 2: Trigonometry.

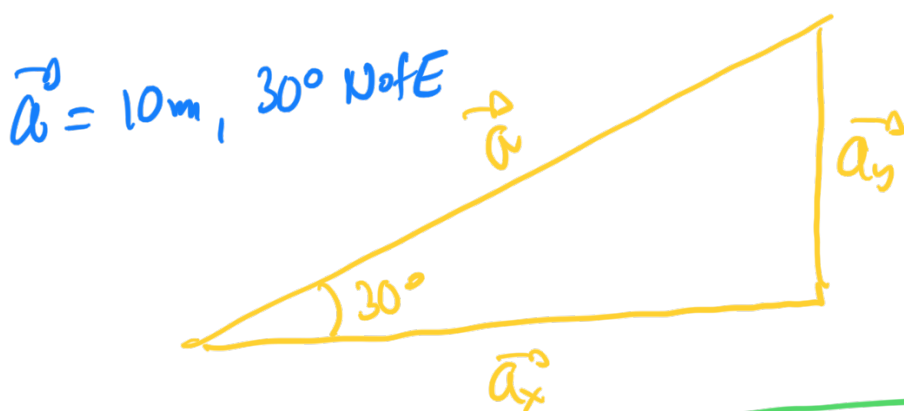
SOH CAH TOA

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\sin 30^\circ = \frac{|\vec{a}_y|}{|\vec{a}|} \quad \cos 30^\circ = \frac{|\vec{a}_x|}{|\vec{a}|}$$

$$\begin{aligned} |\vec{a}_y| &= |\vec{a}| \sin 30^\circ \\ &= (10 \text{ m})(0.500) \\ &= 5 \text{ m} \end{aligned} \quad \begin{aligned} |\vec{a}_x| &= |\vec{a}| \cos 30^\circ \\ &= (10 \text{ m})(0.866) \\ &= 8.66 \text{ m} \end{aligned}$$



$$\vec{a} = 8.66 \text{ m East} + 5.00 \text{ m N}$$

... someone to keep writing the ...

It's common to have units and the directions out all the time. Let's develop a "short hand" notation.

Unit Vectors

Let $\hat{i}$ = 1 m East
 $\hat{j}$ = 1 m North

(Note) $\hat{i}$ = 1 m East
 $\hat{j}$ = 1 m North
 $\hat{i}$ = 1 m East
 $\hat{j}$ = 1 m North
 $\hat{i}$ = 1 m East
 $\hat{j}$ = 1 m North

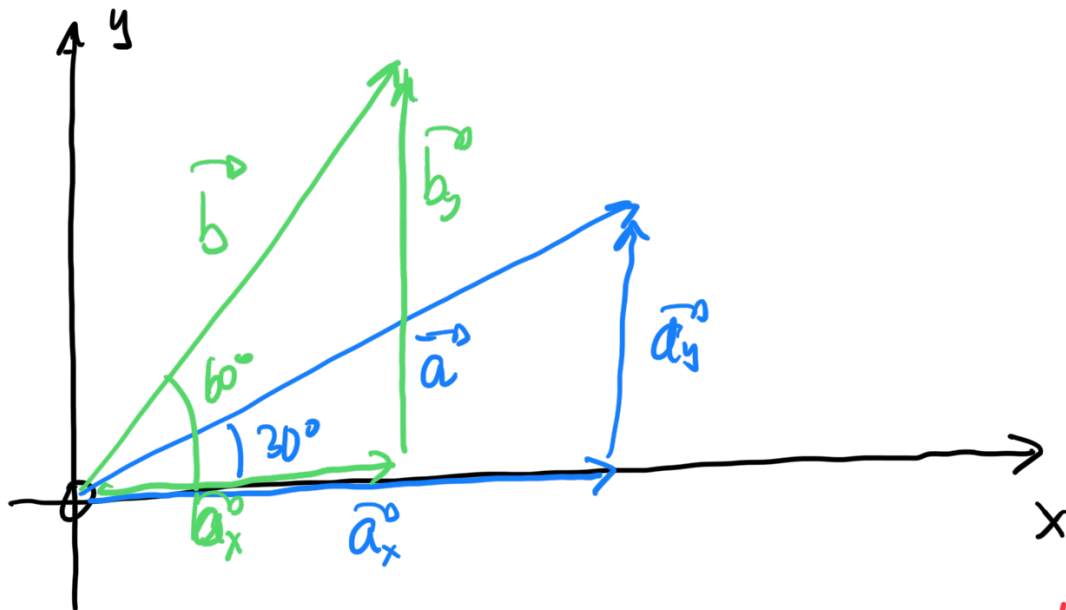
(+ve x-direct)
 (+ve y-direct)

$\vec{a} = 8.66 \hat{i} + 5.00 \hat{j}$

Check: $|\vec{a}|^2 = (8.66)^2 + (5.00)^2$
 $= 75.00 + 25.00$
 $= 100.00$

$|\vec{a}| = 10$





Experiment:

$$\vec{a} = 100\text{ m}, \underline{30^\circ \text{ North East}}$$

$$\vec{b} = 90\text{ m}, \underline{60^\circ \text{ North East}}$$

Step 1: x/y components

$$\vec{a} = \vec{a}_x + \vec{a}_y$$

$$= 100 \cos 30^\circ \hat{i} + 100 \sin 30^\circ \hat{j}$$

$$= 86.60\hat{i} + 50\hat{j} \quad \checkmark$$

$$\vec{b} = \vec{b}_x + \vec{b}_y$$

$$= 90 \cos 60^\circ \hat{i} + 90 \sin 60^\circ \hat{j}$$

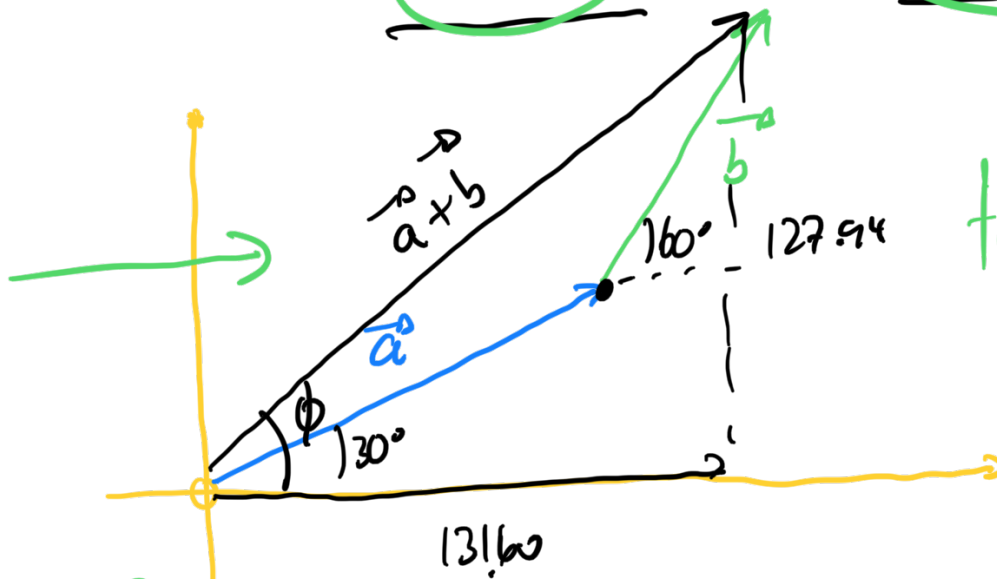
$$= 410 \cos 30^\circ$$

$$= 45 \hat{i} + 77.94 \hat{j} \quad \checkmark$$

Step 2:

What is $\vec{a} + \vec{b}$?

$$\begin{aligned} \vec{a} + \vec{b} &= \left(86.60 \hat{i} + 50 \hat{j} \right) \\ &+ \left(45 \hat{i} + 77.94 \hat{j} \right) \\ &= 131.60 \hat{i} + 127.94 \hat{j} \end{aligned}$$



$$\tan = \frac{\text{opp}}{\text{adj}}$$

Step 3:

$$|\vec{a} + \vec{b}|^2 = 131.60^2 + 127.94^2$$

$$|\vec{a} + \vec{b}| = \sqrt{131.60^2 + 127.94^2}$$

$$\underline{\underline{|a + b|}} = 183.54$$

$$\underline{\underline{\tan \phi}} = \frac{127.94}{131.60} = 0.9722$$

$\tan^{-1}(0.9722)$

$\phi = 44.19^\circ$

∴ $\underline{\underline{a + b}} = 183.54 \text{ m}, 44.19^\circ$
North of East

Subtraction:

This is now easy!

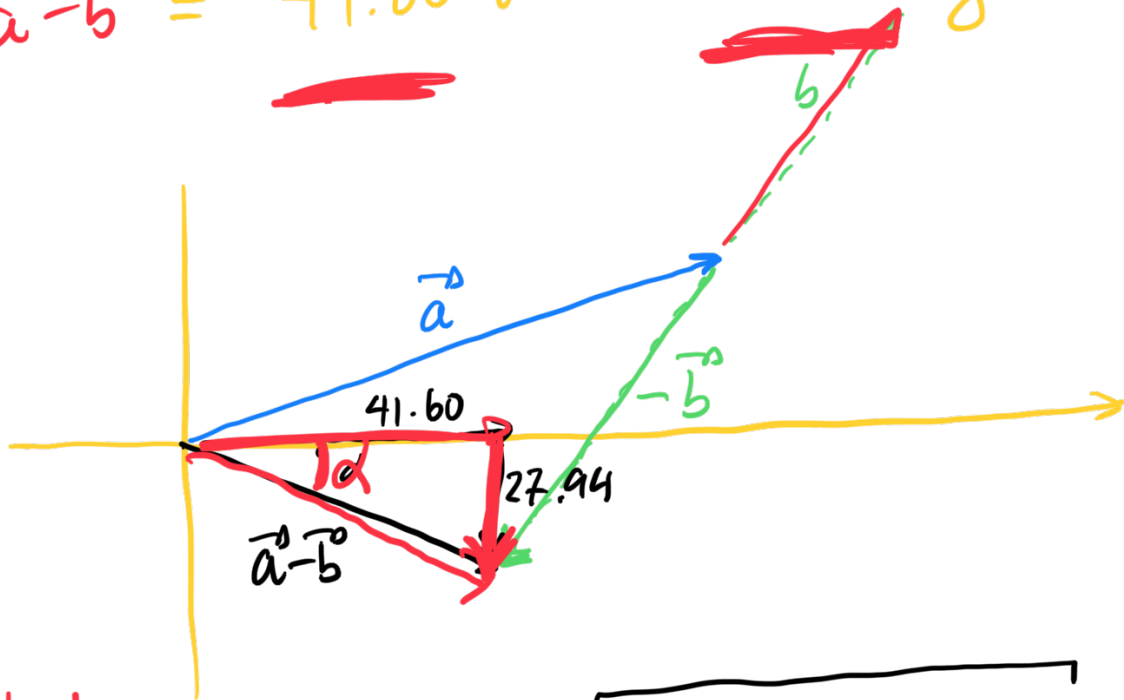
$$\underline{\underline{a - b}} = a + (-b)$$

vector going in
a direction opposite
to b !

$$= 86.60 \hat{i} + 50.00 \hat{j}$$

$$-(45\hat{i} + 77.94\hat{j})$$

$$\vec{a} - \vec{b} = 41.60\hat{i} - 27.94\hat{j}$$



Step 3

$$|\vec{a} - \vec{b}| = \sqrt{41.60^2 + 27.94^2}$$

$$= 50.11 \text{ m}$$

$$\tan \alpha = \frac{27.94}{41.60} = 0.6716$$

$$\alpha = 33.89^\circ$$

$$|\vec{a} - \vec{b}| = 50.11 \text{ m}, \quad 33.89^\circ \text{ South of}$$

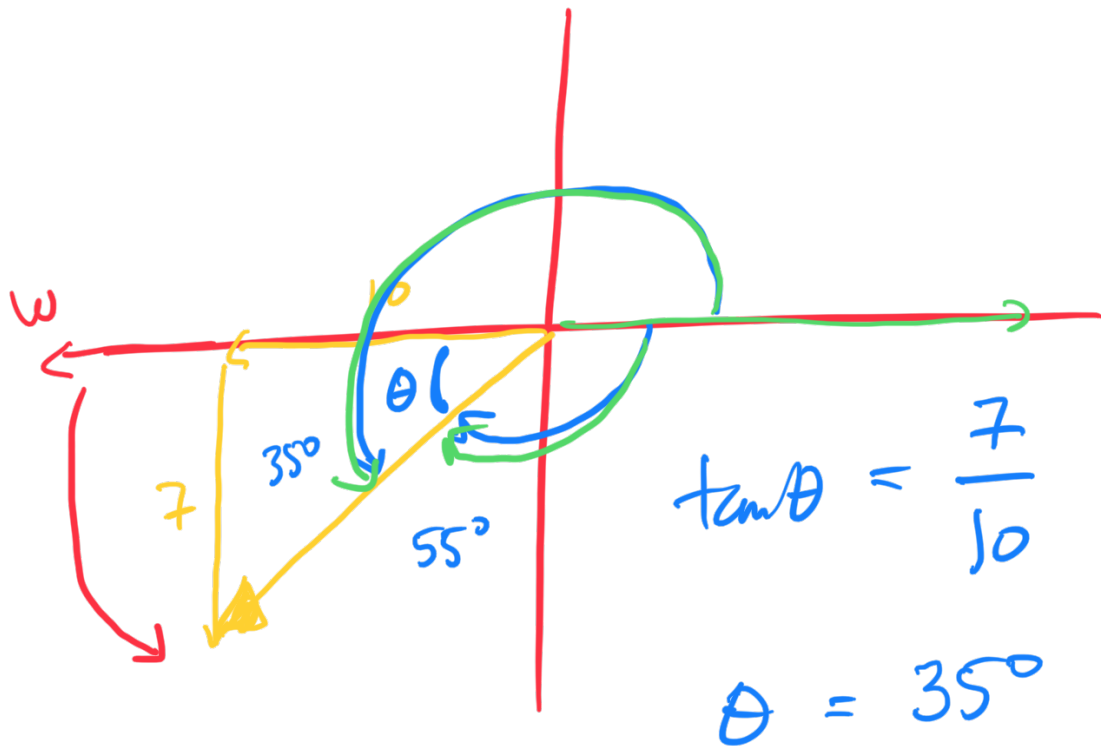
$$|a - b| - - -$$

Fast

Hints for Vector Addition and Subtraction.

- ① Break up all vectors in the problem into x - and y - components.
 - ② Remember Pythagorean Theorem and SOHCAHTOA
 - ③ Draw Pictures!! This is crucial to get the signs of angles correct!!
-

$$\vec{C} = -10\hat{i} - 7\hat{j}$$



✓ 35° South of West

✓ $\theta = \underline{215^\circ}$ $(180^\circ + 35^\circ)$

✓ $\theta = -145^\circ$ $\nearrow$ Not E
 $\nwarrow$ S of E

MM II & D.D. → WAIT

test 1
week 5

15 weeks

Test 2

Exam

$$y = mt + b$$

$[m][t] = L$

↑
T

dimension units

L
m

$t = t^{-1}$

$$C_m = L/T = L$$

v_{m72} m/s

